1. DATASET

Adult Income Dataset (UCI)

- 30,162 records after cleaning
- 14 attributes + target
- Target: Income > 50K (1) or ≤ 50K (0)
- Sensitive Attributes: Sex, Race

ATTRIBUTE CATEGORIES

- Demographic (age, sex, race)
- Socio-economic (education, occupation, marital status)
- Financial (capital gain, capital loss)
- Work-related (hours per week)

RESEARCH QUESTIONS

- **RQ1** How do different ML models perform in terms of fairness and accuracy?
- **RQ2** Which features most influence income prediction?
- **RQ3** How does historical bias in training data influence allocation outcomes across demographic groups?
- **RQ4** Which fairness metric is most appropriate under limited resource allocation?
- **RQ5** Can reinforcement learning allocate resources fairly in dynamic environments?

2. METHODOLOGY WORKFLOW

1 Dataset Acquisition
Adult Income Dataset (UCI)

→ **2 Data Preprocessing**
- Missing value removal
- Data cleaning
- Label encoding
- Feature scaling
- Train/Test split (80/20)

→ **3 Model Development**
- Logistic Regression
- Random Forest
- Prediction Generation

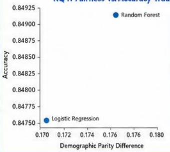
→ **4 Fairness Auditing Framework**
- 4.1 Fairness-Accuracy Tradeoff (RQ1)
- 4.2 Explainability Analysis Feature Importance (RQ2)
- 4.3 Bias Analysis Gender Bias Race Bias (RQ3)
- 4.4 Fairness Metrics DP, EO, Predictive Parity, Accuracy (RQ4)
- 4.5 Resource Allocation Utility vs Fairness Simulation (RQ5)

→ **5 Results Integration & Interpretation**
- Compare models
- Evaluate fairness
- Analyze bias
- Simulate allocation
- Provide insights

→ **6 Recommendations & Guidelines**
Fair and effective resource allocation strategies for real-world deployment

3. RESULTS (RQ1 – RQ5)

RQ1: Fairness vs. Accuracy Trade-off

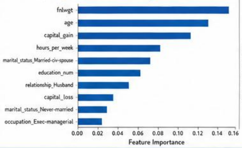


Model Comparison

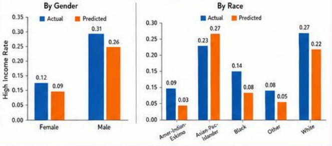
Model	Accuracy	DP Difference (↓)
Logistic Regression	0.8475	0.170
Random Forest	0.8493	0.180

Random Forest achieves higher accuracy with a slightly higher fairness gap.

RQ2: Top 10 Features Influencing Income Prediction

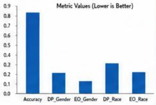


RQ3: Actual vs. Predicted High Income Rates

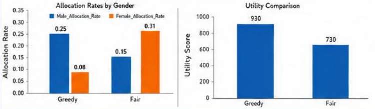


RQ4: Fairness Metric Comparison

Metric	Gender Difference (↓)	Race Difference (↓)
Accuracy	0.85	0.60
Demographic Parity (DP)	0.16	0.23
Equal Opportunity (EO)	0.07	0.06
Predictive Parity (PP)	0.07	0.14



RQ5: Fair Resource Allocation Simulation (20% Allocation Rate)



Simulation Settings
- Total resources: 20% of population
- Strategies compared: Greedy vs Fair (balanced)
- Utility = # of true positives allocated

Fair allocation reduces disparity but incurs moderate utility cost.

4. KEY FINDINGS

- Random Forest provides the best trade-off between accuracy and fairness.
- Key predictors: fnlwgt, age, capital_gain, hours_per_week.
- Significant gender and race disparities exist in predicted outcomes.
- Equal Opportunity shows the lowest disparity among fairness metrics.
- Fair allocation improves equity but slightly reduces overall utility.

5. IMPLICATIONS

- Fairness-aware ML is essential for equitable resource allocation.
- Policy makers should consider multiple fairness metrics.
- Balanced allocation strategies are crucial in constrained settings.
- Explainability helps identify influential factors driving predictions.
- Framework can be extended to other domains and dynamic settings.

6. FUTURE WORK

- Integrate reinforcement learning for adaptive allocation.
- Explore causal fairness and counterfactual analysis.
- Evaluate long-term impact of fair policies.
- Extend to intersectional groups (e.g., gender × race).
- Deploy and validate on real-world systems.

LEGEND

Data · Process · Model · Fairness · Results · Insight

Abbreviations: DP = Demographic Parity | EO = Equal Opportunity | PP = Predictive Parity | Utility = Number of True Positives Allocated

↓ Lower values are better for fairness metrics